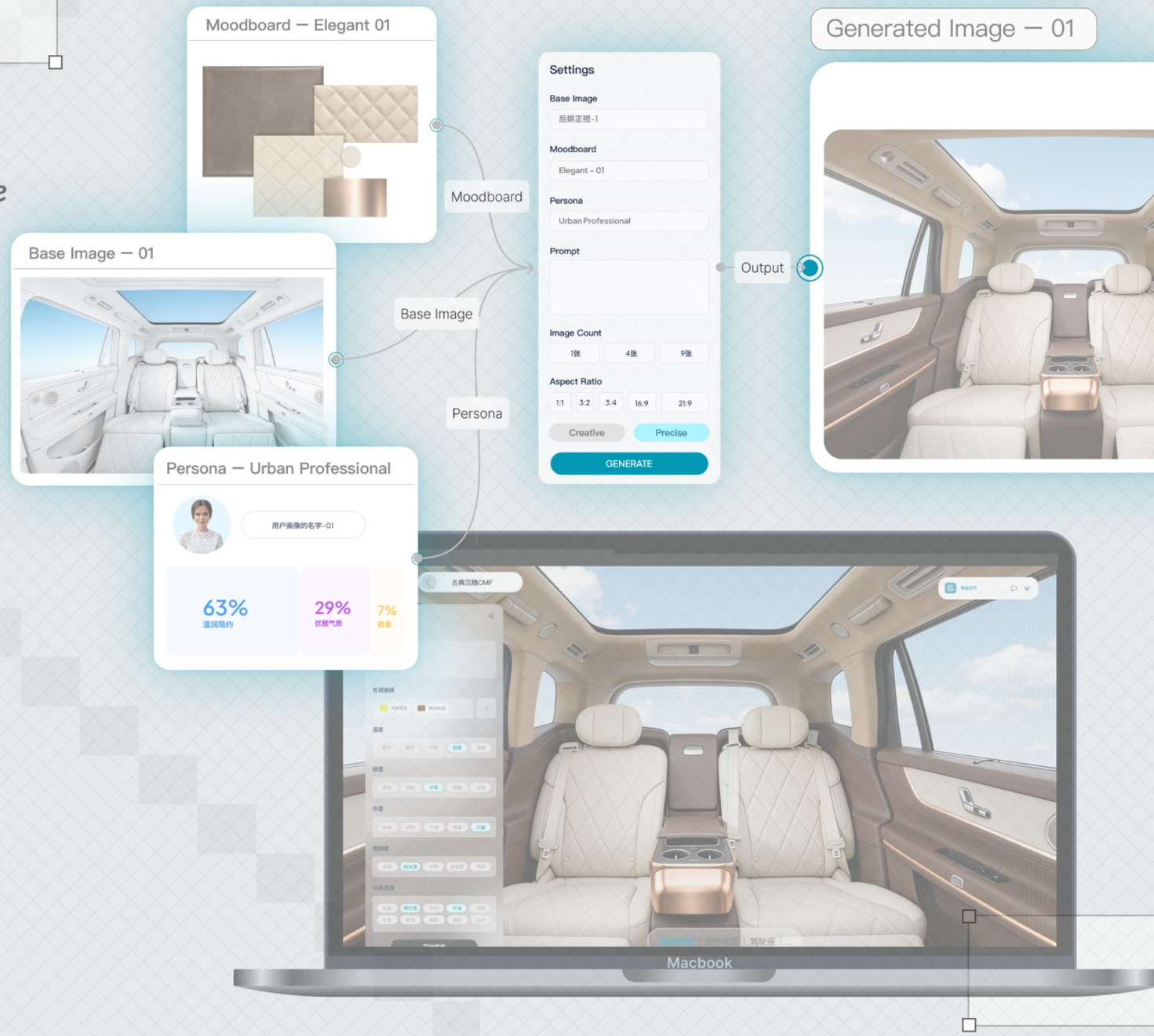
Graduation Project

CulturGen

CMF / Generative AI / Web Design / Human-AI Collaborative

Visit Website:

https://culturgen-fsm.onrender.com



BRIEF

By **integrating cultural elements with generative AI**, I creates an all-in-one **generative design platform** that spans inspiration exploration, material combinations, and CMF concept generation and validation, enabling **automotive designers to efficiently explore** broader design possibilities.

INDIVIDUAL PROJECT

DURATION

Sep. 2025 – Jul. 2026

NEW PROSPECTS

01 From Mobility to Experience

Automotive interior design is shifting **from functionality toward personal experiences**, making CMF a key medium for expressing brand identity and emotion.



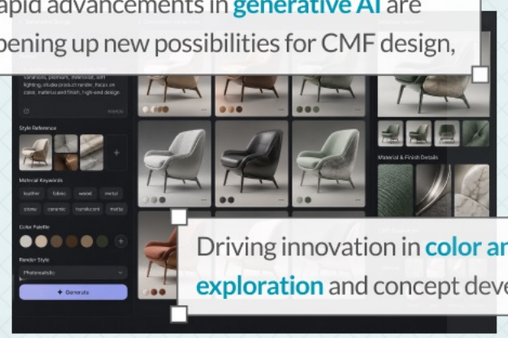
02 Cultural Translation

Chinese-inspired design often relies on superficial symbols, lacking **deeper integration of cultural semantics** into design language.



03 AI-Driven Innovation

Rapid advancements in **generative AI** are opening up new possibilities for CMF design,



Driving innovation in **color and material exploration** and concept development.

03 The Rise of New Energy Vehicle Brands

BYD · XPENG · CHERY · LEAPMOTOR · NIO · GEELY · ZEEKR · 红旗 HONGQI · AION · LYNK&CO · 理想 · VOYAH · SAIC · Xiaomi · AVATR · DENZA · ONVO 乐道 · JAECOO · VETA · ZOTYE · WULING · AITO · BAIC · FOTON · CHANGAN · LEADCAR · GAC · IM 智己汽车 · SKYWELL

The rapid growth of **NEV brands** makes **distinctive CMF identities** increasingly important in automotive interior design.

DESK RESEARCH

Challenges in Current CMF Design Practices

Where do existing design approaches fall short? An examination of workflows, and existing AI tools.

Time-Consuming Research
Extensive time spent collecting design references.

Repeated Revisions
Frequent adjustments prolong the design process.

Slow Concept Exploration
Limited efficiency in exploring design alternatives.

Design Process



Time-Consuming CMF Workflow

Lack of Innovation
Overreliance on conventional CMF solutions.

Weak Brand Identity
Difficulty establishing distinctive visual identities.



Limited Design Differentiation

Design Output

Tool System

Multiple Software Tools
Different design tasks require separate applications.

Poor Interoperability & Frequent Switching
Limited compatibility requires constant transitions between tools.



Lack of Tool Integration

Similar Color Palettes
Repetitive color choices across automotive brands.

Fragmented Workflow
Disconnected design processes disrupt workflow continuity, increase complexity, and reduce efficiency.

Traditional Automotive CMF Design Workflow

Where do existing design approaches fall short? An examination of workflows, and existing AI tools.



01 Inspiration & Trend Research

Explore inspiration and trends to define innovative CMF directions.



03 Digital CMF Visualization

Visualize schemes in automotive interiors through digital material and lighting simulation.



05 Prototyping & Validation

Validate material and visual qualities of CMF schemes using physical mock-ups.



02 Scheme Development

Combine colors, materials, and finishes to develop diverse CMF schemes.



04 Design Review & Refinement

Compare and evaluate CMF schemes to identify refinement opportunities.

OPPORTUNITY

Traditional automotive CMF design is an iterative process integrating cultural, brand, and material insights. Its reliance on human expertise for inspiration translation and design development **creates opportunities for AI-assisted exploration.**

INTERVIEW



■ 01 From Mobility to Experience

Semi-structured interviews *identified user needs and pain points, while* ***prototype testing interviews*** *gathered feedback.*



Miss. Wang

Automotive CMF design engineer

Five years of experience, proficient in design software and AI image-generation tools.

> AI-generated images often suffer from **structural and material distortions**, while time-consuming visualization **reduce design efficiency.**

Needs: Logically consistent CMF material details

Industrial Design Undergraduate

Experienced in CMF of transportation design competitions, familiarity with AIGC tools.

> AI-generated **lack realism and precise control**, while collecting images and text across **multiple platforms** makes the workflow complex.

Needs: Precise control; Integrated Design Tool



Miss. Dong

Interviews face limited AI control, inaccurate CMF representation, and fragmented workflows, seeking precise control and efficient iteration.

ANALYSIS

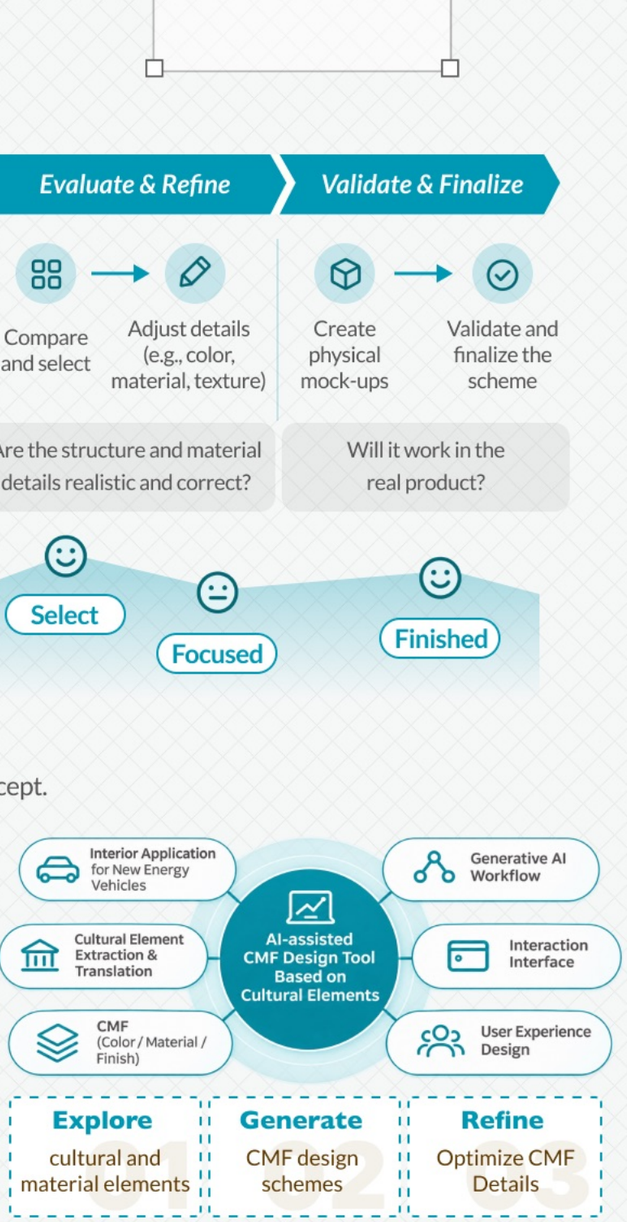
■ User Journey Map

Stages	Inspiration & Research	Prompt & Generate	Evaluate & Refine	Validate & Finalize
Actions	Collect trend references → Analyze CMF materials and styles	Set prompts and CMF parameters → Generate scheme options	Compare and select → Adjust details (e.g., color, material, texture)	Create physical mock-ups → Validate and finalize the scheme
Thinking	What are the relevant trends and CMF directions?	Can I quickly generate diverse and relevant options?	Are the structure and material details realistic and correct?	Will it work in the real product?
Feeling	Curious	Exciting	Bother	Select → Focused	Finished

■ Ideation

Translating user insights into design opportunities and a clear product concept.

Painpoint	Scattered cultural references; Limited control over AI generation; Inaccurate CMF material and structural details; Iteration is time-consuming
How Might We	How might we effectively translate cultural elements into CMF design? How might we make AI generation more controllable and efficient?
Opportunity	Cultural Resource Library; Precise CMF Control; AI-assisted Generation; Integrated Workflow



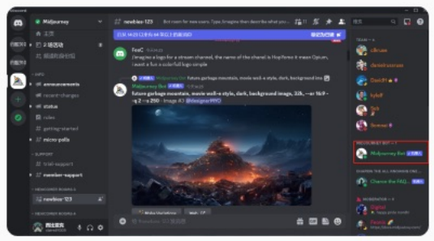
EXPLORATION

■ AI Generation Interaction Study

Where do existing design approaches fall short? An examination of workflows, and existing AI tools.

01 Prompt-based Generation

Ref. Midjourney



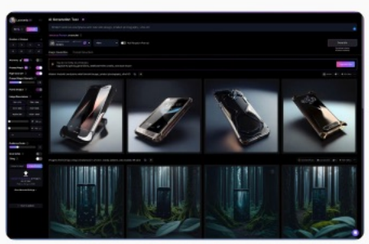
Generate and refine images through text prompts and visual references.

Rapid concept exploration. Low-barrier inputs.

Limited control over details. Local modifications.

02 Adobe Firefly — Parameter-based

Ref. Leonardo AI



Adjust styles, references, and visual attributes through parameter controls.

Visual controls enable intuitive adjustments, predictable iteration.

Limited support for professional, material-specific parameters.

03 Node-based Generation

Ref. ComfyUI



Build customizable generation workflows by connecting functional nodes.

Flexible workflows offer granular control and reusable processes.

Complex node structures increase learning and operation costs.

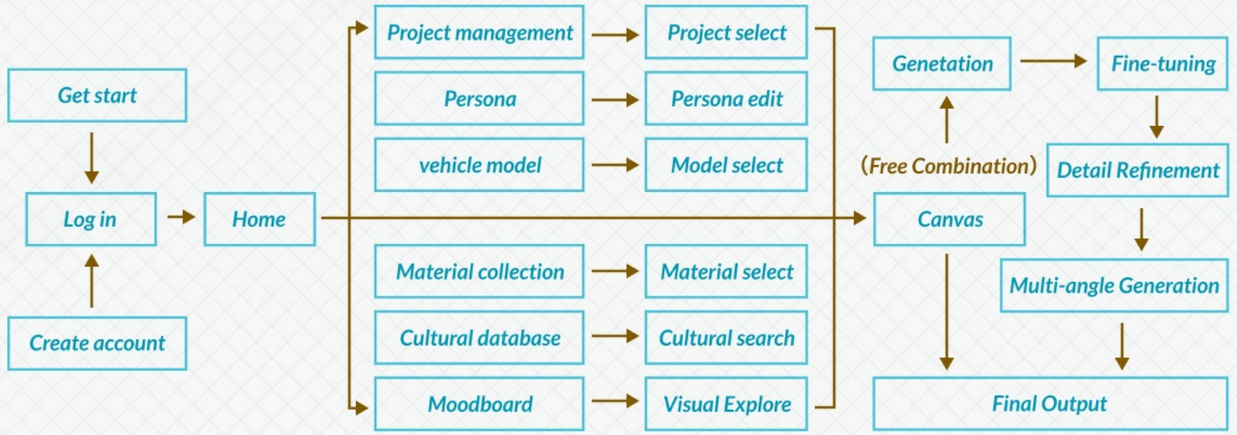
UI DESIGN

■ Web Structure

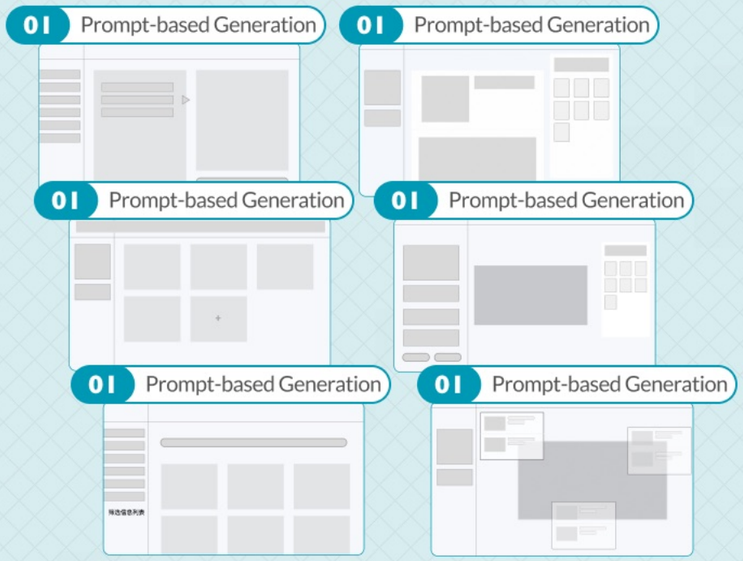
- **Home**
  - **Project management**
    - *Project management*
      - Create/ Open/ Manage
    - *User persona*
      - Description
      - Age Group
      - Scenario
    - *Vehicle library*
      - Multi-angle Images
  - **Settings**
    - *Preferences*
      - Theme mode
    - *profolio*
- **Resource Exploration**
  - **Database**
    - *Filter by artifact*
      - Artifact list
    - *Filter by material*
      - Material list
    - *Material details*
      - Style/ Color categories
      - Source
      - AI Suggestion
  - **Material collection**
    - *My material*
      - Browse/ Save/ Select
    - *Moodboard*
      - Search
      - Upload
      - Combine/ Edit
- **Canvas**
  - **Canvas creation**
    - *Select node*
      - Base model
      - Moodboard
      - User persona
      - API Generation
      - Generation Settings
      - Reference Images
    - *AI assistant*
      - Prompt optimization
  - **Fine-tuning**
    - *Local Refinement*
      - Parameter adjustment
      - Lighting adjustment
      - Texture Adjustment
    - *Multi-angle Generation*

■ User Flow

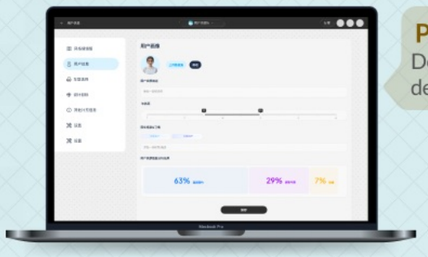
How users interact with the product, step by step.



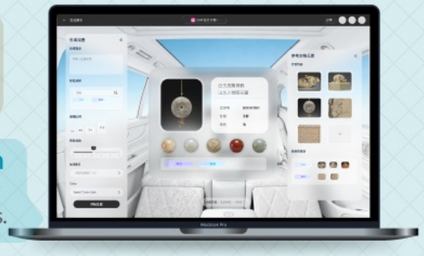
Wireframes



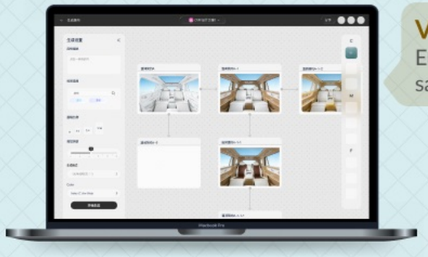
High-Fidelity UI Design



Persona setup
Define vehicle models and target users to align design preferences, improve generation accuracy.



Cultural Element Integration
Display cultural elements in panels, allowing users to explore, adjust, and preview designs.



Version Management
Enable collaboration on a shared canvas, with saved versions for easy tracking and comparison.



Multi-angle Generation
Combine images, materials, and multi-angle base models to generate consistent CMF designs.

Testing & Iteration

Built an interactive prototype to test user flows, gather feedback on usability issues and refine the design based on user insights.

■ UI Design System

01 LAYOUT & GRID

1920 x 1080
12-column grid



Sidebar
339pt

Header
96pt

Fixed navigation
Main content
Modal overlay

02 UI COLORS

Primary	Hover
#039FB6	#028EA3
Light	Deep teal
#EAFBFE	#063E46
Header	Assist
#111111	#93DDF6

03 TYPOGRAPHY

Aa Lato

Regular 400 / Bold 700

H1	32 / 48	Bold
H2	24 / 36	Bold
H3	20 / 30	Bold
Body	14 / 22	Regular

04 RADIUS

- 6px — Tags
- 8px — Controls
- 12px — Cards

05 CORE COMPONENTS

Generate | Cancel | Ceramic

Search materials | Material

Personal | Team | 1 | 2 | 3

Cards / Tables / Pagination / Forms

WEB DESIGN

01 User Persona

Explore user segments and preferences, and easily create or refine personas to **meet individual needs**.

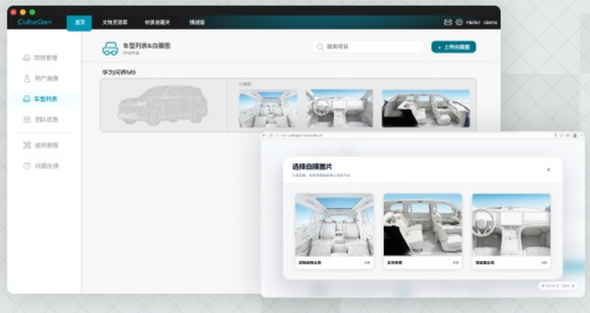
User Persona List | Add User Persona



AI Assistant Analysis

02 Vehicle library

Easily browse vehicle models and their multi-angle base images, and quickly **add or update vehicle data for efficient design generation**.

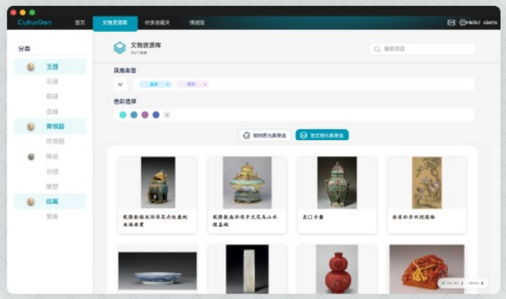


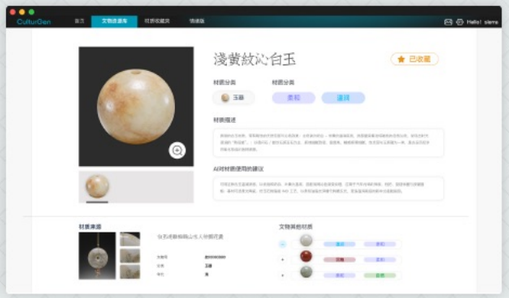
Upload & Collect Model | Add in canvas

03 Artifact & Material Library

Easily browse and categorize artifacts and materials, quickly edit artifact information and material properties, and efficiently **manage and access resources.**

Browse | Filter by Tags | Inspiration | Save & Favorites | Material Details

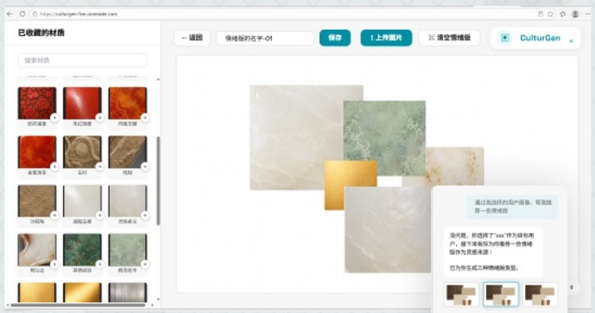




01 Moodboard

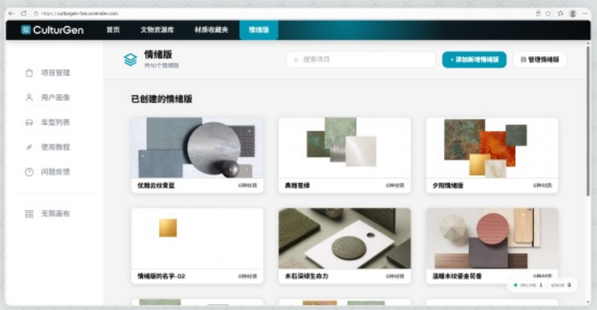
Select and freely **combine materials** to create moodboards, quickly develop or **refine concepts for design generation**.

Moodboard Creation | Creative Exploration



AI-Powered Recommendations

Moodboard Management



Core Function — Infinite Canvas

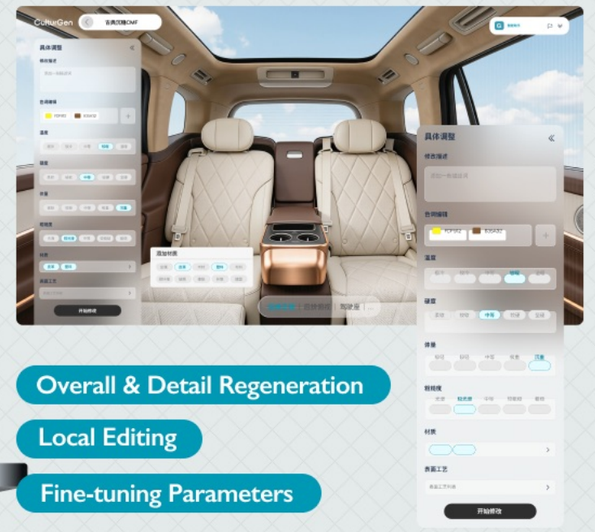


ComfyUI Workflow | API Integration | LLM Agent



01 Moodboard

Allowing users to explore and enhance concepts while maintaining visual consistency and creative freedom.



Overall & Detail Regeneration

Local Editing

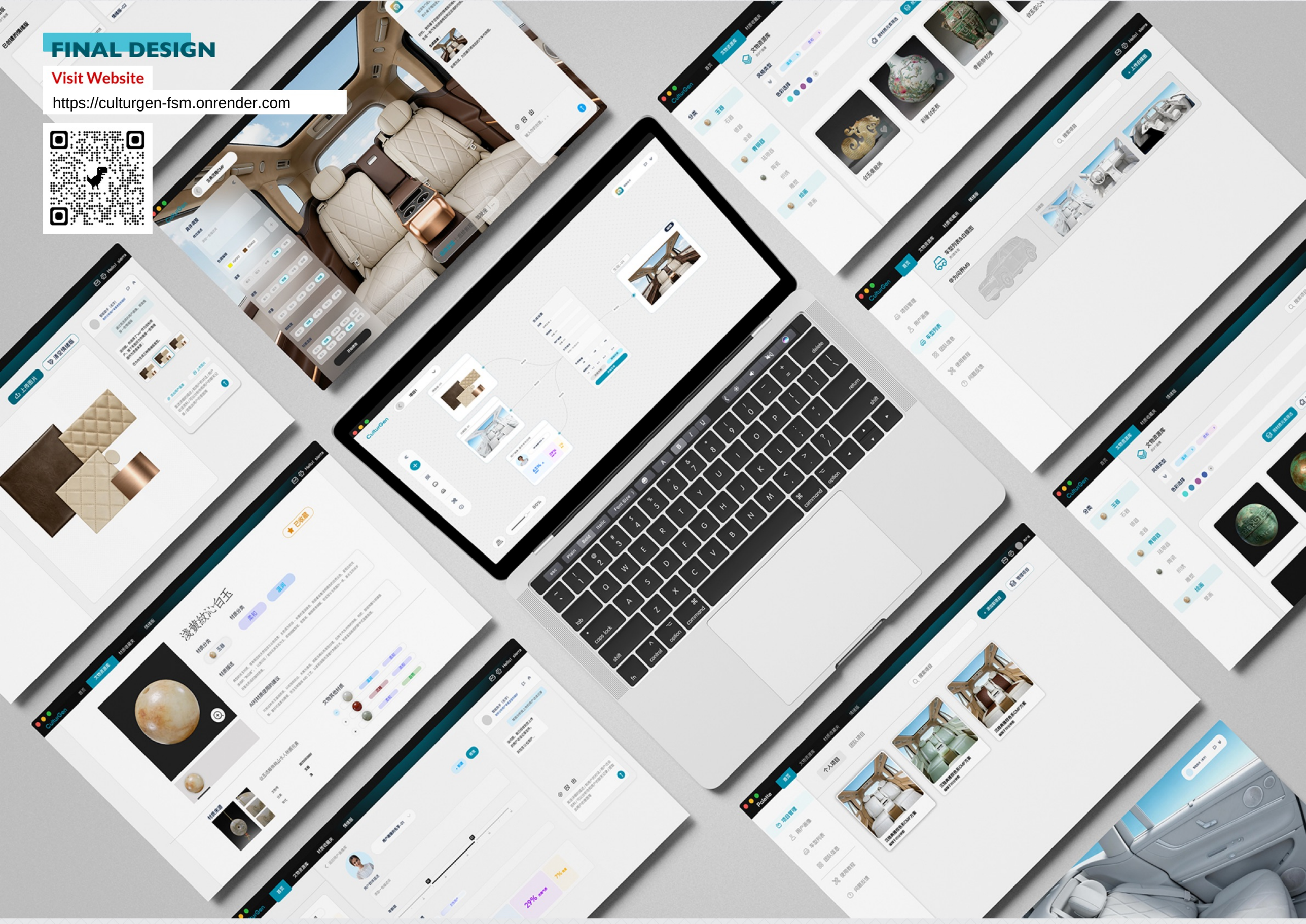
Fine-tuning Parameters

Canvas of CulturGen Offers three AI-assisted approaches to CMF design generation, allowing users to freely combine **base models, user personas, moodboards, images, and text** as references for flexible creative exploration.

FINAL DESIGN

Visit Website

https://culturgen-fsm.onrender.com



IMPLEMENTATION

Web Development & AI Integration

■ System Architecture

User layer

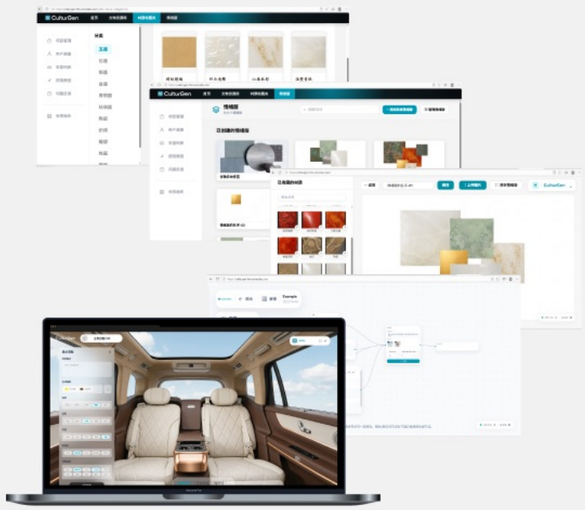
Designer

Researcher



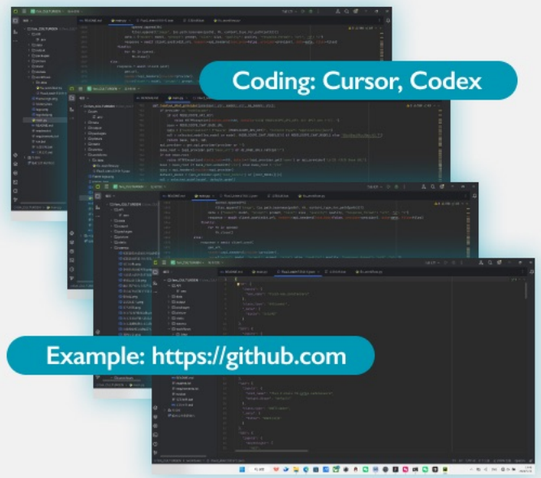
Frontend

Web application



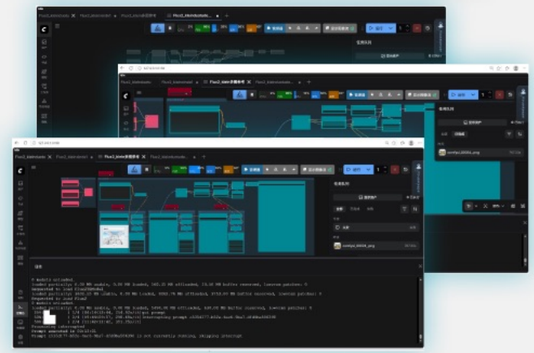
Backend **Cursor**

Developed the web application using Python, HTML, CSS, and JavaScript.



Generate workflow

Locally deployed customizable ComfyUI workflows



with cloud-based image generation via AI model APIs.

■ Experience Showcase









REFLECTION

The current process offers limited control over specific CMF details, such as material textures and color combinations...

We can Introduce adjustable CMF parameters and localized editing tools to enable more precise and flexible design refinement.